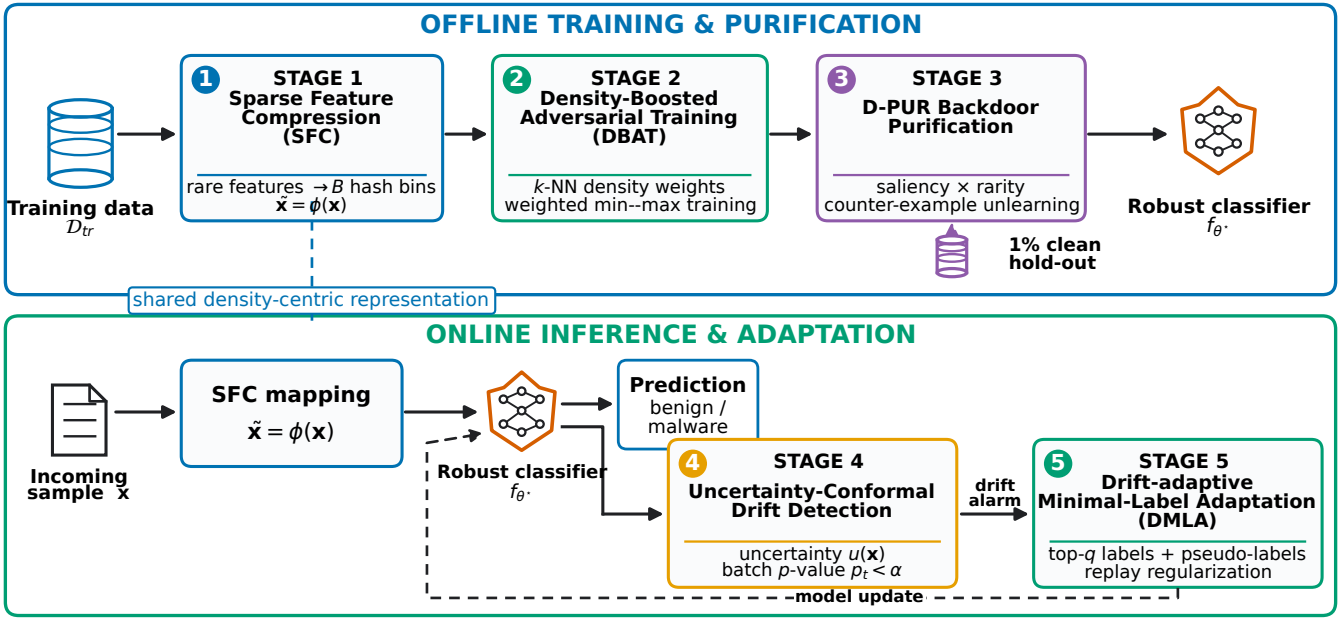
OFFLINE TRAINING & PURIFICATION
Training data
$\mathcal{D}_{tr}$
1
STAGE 1
Sparse Feature Compression (SFC)
rare features $\rightarrow B$ hash bins
$\tilde{\mathbf{x}} = \phi(\mathbf{x})$
2
STAGE 2
Density-Boosted Adversarial Training (DBAT)
k-NN density weights
weighted min--max training
3
STAGE 3
D-PUR Backdoor Purification
saliency × rarity
counter-example unlearning
1% clean hold-out
Robust classifier
f_θ.
shared density-centric representation
ONLINE INFERENCE & ADAPTATION
Incoming sample $\mathbf{x}$
SFC mapping
$\tilde{\mathbf{x}} = \phi(\mathbf{x})$
Robust classifier
f_θ.
Prediction
benign / malware
4
STAGE 4
Uncertainty-Conformal Drift Detection
uncertainty $u(\mathbf{x})$
batch p-value $p_t < \alpha$
drift alarm
5
STAGE 5
Drift-adaptive Minimal-Label Adaptation (DMLA)
top-q labels + pseudo-labels
replay regularization
model update